



HOME / POWER ELECTRONICS / CUSTOM TRANSFORMERS

[CUSTOM MAGNETICS](#) / [SMPS DEVELOPMENT](#)

Custom high-frequency transformers *built for the next bench test.*

Herder Elektronische Systemen develops custom transformer prototypes for switch-mode power supplies, connecting electrical requirements, core selection, winding construction, 3D documentation and measured bench behavior.

[DISCUSS A TRANSFORMER PROTOTYPE →](#)

[VIEW THE 5 V / 1 A SMPS PROJECT →](#)

[FLYBACK](#)

[SMPS MAGNETICS](#)

[CUSTOM WINDING](#)

[3D MODELING](#)

[PROTOTYPE VALIDATION](#)

Magnetics shaped *around the converter.*

- Flyback transformer prototypes
- High-frequency isolation transformers
- Custom turns ratios
- Primary, secondary and auxiliary windings
- Gapped ferrite-core prototypes
- RM, EE, EFD, ETD and other practical ferrite geometries where suitable
- Hand-wound engineering samples
- Small prototype quantities
- Revisions of existing transformer designs
- Replacement prototypes where sufficient source information is available

Feasibility depends on topology, input range, output requirements, switching frequency, isolation needs, core geometry, winding window, thermal limits, safety expectations and component availability.

Bring the electrical *boundary.*

REQUIREMENTS PANEL

Start with what is known.

Converter topology

Input-voltage range

Output voltage and current

Switching frequency

Controller or primary-side IC

Target power

Number of windings

Auxiliary-bias requirements

Isolation requirement

Core or size constraints

Expected duty cycle

Existing schematic

Existing transformer data

Target quantity

Intended environment
Safety or qualification expectations

Do not send sensitive design files through the initial email.

Secure file exchange can be arranged after the first discussion.

From requirements *to a reviewable prototype.*

01

Define the electrical boundary

Confirm topology, voltage, current, frequency, duty cycle and isolation needs.

02

Select the magnetic structure

Review core material, geometry, effective area, winding window and gap approach.

03

Develop the winding construction

Define turns, wire, winding order, insulation assumptions, polarity and pin assignment.

04

Model and document the assembly

Produce construction drawings, winding-stack visuals and 3D reference models where included.

05

Wind and inspect the prototype

Build the engineering sample and record construction details.

Measure and revise

Review inductance, DCR, leakage, switching behavior and converter-level observations where test access permits.

3D models that make the construction *reviewable*.

Parametric models can make transformer construction, pin orientation, winding space and mechanical integration easier to inspect before or after the first physical build.

- Parametric modeling of cores, bobbins, pins and assemblies
- Simplified winding-layer models
- Insulation-stack visualization
- Core-gap representation
- Winding-window utilization views
- Exploded assembly views
- Pin numbering and polarity orientation
- PCB-mounted clearance review
- Enclosure and neighboring-component clearance review
- Dimensioned reference drawings
- Technical renders for documentation

POTENTIAL CAD DELIVERABLES

Native parametric model

STEP export

Dimensioned drawing

Exploded view

Winding-stack illustration

Pinout and polarity diagram

PCB clearance view

The 3D model documents physical construction and integration.

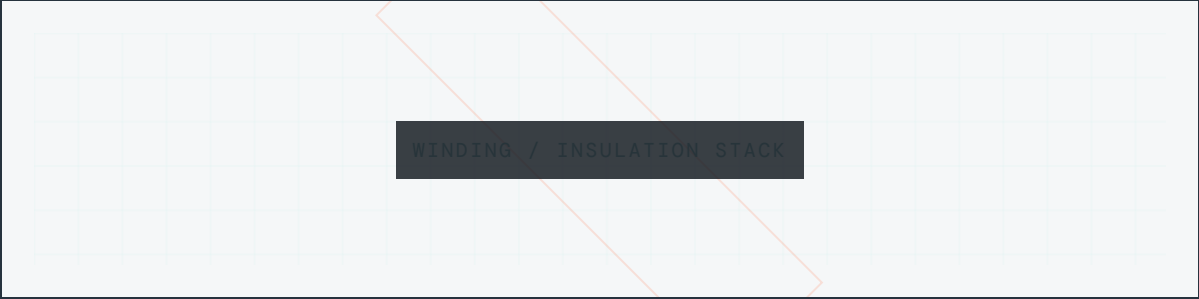
It does not by itself validate electromagnetic performance, insulation coordination, thermal behavior, regulatory compliance, manufacturability or production repeatability.



Complete transformer assembly



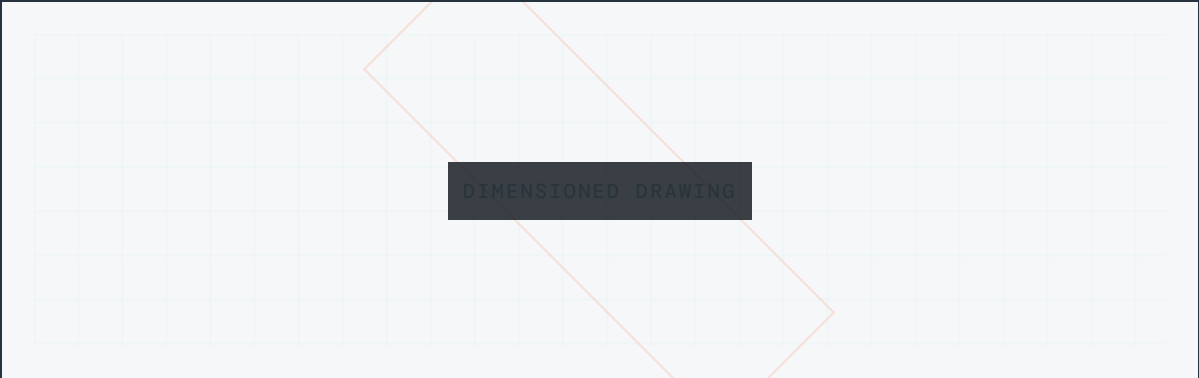
Exploded core and bobbin view



Winding and insulation stack



PCB-mounted clearance view



Dimensioned reference drawing

A prototype with enough *information to continue.*

- Wound transformer prototype
- Winding specification
- Turns table
- Wire and insulation notes
- Core and gap information
- Pinout and polarity
- Inductance measurements
- DCR measurements
- Leakage-inductance observations
- Construction notes
- 3D construction model where included
- Bench-test notes
- Revision recommendations

Exact deliverables depend on the agreed scope, available test access and whether the request covers transformer construction only or converter-level evaluation.

Measurements with the *test conditions attached.*

TRANSFORMER PHOTOGRAPH

Caption / provenance / revision

WINDING DIAGRAM

Turns, polarity and pin assignment

CORE / GAP DETAIL

Core geometry and gap detail

SCHEMATIC EXCERPT

Transformer in converter context

INDUCTANCE MEASUREMENT

Instrument / date / test condition

DCR MEASUREMENT

Instrument / date / winding

LEAKAGE MEASUREMENT

Winding condition and method

SWITCHING WAVEFORM

Operating point / probe / revision

LOAD-TEST RESULT

Input, load and thermal context

3D ASSEMBLY MODEL

Model revision and CAD provenance

Every evidence block should identify source or provenance, instrument, test date, operating condition and prototype revision. No values, plots, dimensions, photographs or validation claims are invented here.

Prototype magnetics are engineering samples.

Suitability for production, reinforced insulation, regulatory compliance, long-term reliability, thermal performance and manufacturing repeatability requires separate analysis, testing and qualification.

07 / RELATED WORK

See the transformer *in converter context.*

PUBLIC TECHNICAL PROTOTYPE

5 V / 1 A SMPS

Offline flyback prototype with custom transformer work, schematics, waveform captures and bench-validation notes.

[VIEW PROJECT →](#)

[VIEW REPOSITORY ↗](#)

[POWER ELECTRONICS PROTOTYPING →](#)

○ [START A TECHNICAL DISCUSSION](#)

Bring the *converter* *requirements.*

Share the topology, electrical targets, switching frequency, intended quantity and what has already been measured. Secure file exchange can follow the initial discussion.

[DISCUSS A TRANSFORMER PROTOTYPE →](#)

Or write directly: info@herdersystemen.com



HERDER
ELEKTRONISCHE SYSTEMEN

© 2026 Herder Elektronische Systemen

[Power electronics](#) [Privacy](#) [Engagement notes](#) [Back to top ↑](#)